Gianni Pilotti

ECONOMICS & FINANCE · PORTFOLIO MANAGEMENT · MARKET RISK · QUANTITATIVE FINANCE · 6-MONTH INTERNSHIP FROM SEP 2026

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PROFILE

Final-year Bachelor student in Economics & Finance at the University of Luxembourg with a hands-on focus on quantitative finance. Built Python models spanning portfolio optimisation, VaR & stress-testing, multi-factor equity strategies, options pricing and on-chain market microstructure. AMF & Bloomberg certified. Targeting a 6-month internship in portfolio management, market risk or quantitative finance from September 2026.

EDUCATION

University of Luxembourg

Bachelor in Economics and Management — Finance

Esch-sur-Alzette · Sep 2023 – Jan 2027 (Expected)

Financial Markets · Corporate Finance · Macroeconomics · Econometrics · Data Analysis

University of Economics in Bratislava (EUBA)

Erasmus+ Exchange Semester

Bratislava, Slovakia · Sep 2025 – Jan 2026

Quantitative Methods · Public Finance

QUANTITATIVE PROJECTS

Portfolio Optimisation — Markowitz, Black-Litterman & Risk Parity [Portfolio]

– Built a mean-variance optimiser (efficient frontier, max-Sharpe, min-variance) and a Black-Litterman model blending market equilibrium with investor views on a 10-stock universe.

– Implemented a Risk Parity allocator; max-Sharpe portfolio reached Sharpe 1.37 vs 0.79 for min-variance.

Python · NumPy · pandas · SciPy · Matplotlib · yfinance

VaR & CVaR — Portfolio Risk & Stress-Testing [Risk]

– Estimated 99% VaR/CVaR on a 9-asset multi-region portfolio via Historical Simulation, Parametric variance-covariance and Monte Carlo with 50,000 Cholesky-correlated paths.

– Stress-tested against 2008 crisis, 2020 COVID crash and 2022 rate-hike selloff; Parametric VaR shown to understate tail risk (CVaR 99% −13.24% on BTC vs −4.58% diversified).

Python · NumPy · pandas · SciPy · yfinance · Plotly · Matplotlib

Multi-Factor Equity Strategy — Backtest vs S&P 500 (2016–2026) [Portfolio · Quant]

– Designed a long-only Momentum/Value/Quality composite score, backtested monthly on a 25-stock universe; outperformed SPY: Sharpe 1.38 vs 0.97, annualised return 28.9% vs 15.1%.

Python · NumPy · pandas · Matplotlib · yfinance

Options Pricing Engine — Black-Scholes & Binomial Tree (CRR) [Risk · Quant]

– OOP engine covering European and American options with four Greeks (Δ, Γ, ν, Θ), dividend adjustment; verified binomial convergence to Black-Scholes and quantified early-exercise premium.

Python · NumPy · SciPy · Plotly · ipywidgets

On-Chain Market Microstructure — Uniswap V4 / DeFi [Quant]

– Extracted live pool state from Uniswap V4's Singleton PoolManager via raw storage reads (extsload); estimated Kyle's lambda on 6,737 real ETH/USDC swaps ($t = 317$, $R^2 = 0.94$) — realised vol 24%, effective spread 10 bps.

Python · NumPy · pandas · web3 · eth-abi · on-chain data · econometric estimation

PROFESSIONAL EXPERIENCE

Proximus Luxembourg Bertrange, Luxembourg · Jul – Aug 2026

Summer Student — IT Systems & Infrastructure

– Managed IT asset tracking and inventory across the department, ensuring data accuracy in internal systems.
– Supported workstation deployment, telephony configuration and server-side tasks within the systems engineering team.

ONLINE SIMULATIONS

Risk Management Simulation
Goldman Sachs · Forage
Aug 2026
VaR computation, model validation and risk reporting in a sell-side environment.

Investment Management Simulation
Fidelity International · Forage
Aug 2026
Fund analysis, portfolio construction and client reporting workflows.

Markets Sales & Trading Simulation
Citi · Forage
Aug 2026
Pricing, client advisory and trade execution in a global markets context.

SKILLS

Quantitative Finance	VaR · CVaR / Expected Shortfall · Monte Carlo · Cholesky Decomposition · Historical & Parametric Simulation · Stress Testing · Backtesting · Portfolio Optimisation · Markowitz · Black-Litterman · Risk Parity · Factor Investing · Black-Scholes · Binomial Tree (CRR) · Option Greeks · Market Microstructure · Kyle's Lambda · DCF · WACC · EV/EBITDA
Programming & Data	Python (NumPy · pandas · SciPy · Matplotlib · Plotly · yfinance · web3 · eth-abi) · OOP · SQL · Excel · VBA · Stata · Linear Regression · on-chain data

CERTIFICATIONS

AMF / Bloomberg Market Concepts (BMC) / Bloomberg Finance Fundamentals (BFF) / Bloomberg Spreadsheet Analysis (BSA) / Bloomberg ESG / LSEG Workspace

LANGUAGES & INTERESTS

French Native **English** B2/C1 **German** A2

Macroeconomics / financial markets / geopolitics·/ travels / competitive esports (Rainbow Six Siege — in-game leader)